

Tap into Reality: Understanding the Impact of Interactions on Presence and Reaction Time in Mixed Reality

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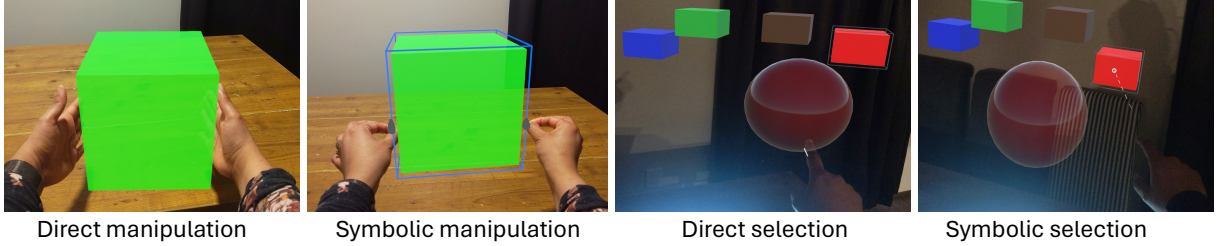


Figure 1: View of the virtual scene for four conditions in MR. In the direct manipulation (DM) condition (left), a box is grabbed with both hands. An unnatural two-hand grab gesture with a hook is used in the symbolic manipulation (SM) condition (center-left). For the direct selection (DS) condition (center-right), a bubble is popped by touch, while in the symbolic selection (SS) condition (right), an air-tap with the ray is used to interact with the box.

Abstract

Enhancing presence in mixed reality (MR) relies on precise measurement and quantification. While presence has traditionally been measured through subjective questionnaires, recent research links presence with objective metrics like reaction time. Past studies examined this correlation with varying technical factors (object realism and behavior) and human conditioning, but the impact of interaction remains unclear. To answer this question, we conducted a within-subjects study ($N = 50$) to explore the correlation between presence and reaction time across two interaction scenarios (*direct* and *symbolic*) with two tasks (*selection* and *manipulation*). We found that presence scores and reaction times are correlated (correlation coefficient of -0.54), suggesting that the impact of interaction on reaction time correlates with its effect on presence.

Index Terms: Mixed Reality, Presence, Interaction

1 Introduction

In Mixed Reality (MR), creating a sense of presence is essential to providing an immersive and engaging experience. Presence, the psychological sensation of *being there*, allows users to feel fully absorbed in a virtual environment (VE) [50]. Measuring presence is essential for improving it, and traditional approaches have primarily relied on subjective questionnaires. However, recent research has identified correlations between presence and objective human performance measures, such as reaction time [12, 13]. These correlations have been mainly explored when presence is influenced by technical factors (appearance and behavior of objects) [71, 57] or by conditioning-related human factors [29, 68].

While these findings have provided valuable insights into the nature of presence, prior work often assumes that specific interaction mechanisms do not affect the relationship between presence and reaction time. However, literature [65, 40, 22, 66, 32, 35], has suggested that interaction design itself influences presence. In MR, *interaction refers to the ways in which users engage and manipulate the VE*. It encompasses the mechanisms and interfaces that allow users to perceive, navigate, and affect the virtual world. Biocca’s work on interactions in VE [9] suggests that users build a mental model of interaction patterns, and any deviation from expected behavior can affect presence.

Similarly, expectations about the realism and consistency of interactions are shown to be essential in establishing a presence in

MR environments [45, 31, 29]. For example, natural interactions may lead to greater presence, while unfamiliar or inconsistent interactions can disrupt presence [36, 55]. On the other hand, *reaction time is typically measured as a response to a cue*, often occurring as an interaction. Therefore, any approach to measuring presence should be sensitive to interactions, as it directly influences reaction time and impacts presence [52].

To address this gap in the literature, we conducted an exploratory within-subjects study ($N = 50$). We evaluated the relationship between presence and reaction time across two interaction mechanisms: *direct* (D) and *symbolic* (S). Direct interaction closely mimics real-world actions, like reaching out to grab or touch a virtual object. In contrast, symbolic interaction uses abstract mechanisms, such as pressing a button or performing a gesture that does not replicate real-world actions but achieves the intended effect. We examined two tasks, *manipulation* and *selection*, to capture a broader range of interactions. The manipulation task involves manipulating virtual objects directly, whereas the selection task demands precise coordination between user intentions and the virtual object.

We followed a 2 (interaction: direct vs. symbolic) $\times 2$ (tasks: manipulation vs. selection) study design to create four conditions: *direct manipulate* (DM), *symbolic manipulate* (SM), *direct select* (DS), *symbolic select* (SS). We used post-experience questionnaires (PQ [71] and IPQ [52]) to measure the change in presence between conditions and systematically measured the reaction time of users in response to a visual stimulus. Based on the collected data, we aimed to answer the following research questions (RQ):

- RQ1:** How does the choice of interaction impact presence?
- RQ2:** Does the nature of the task to be performed impact the effect of interaction on presence?
- RQ3:** Does the correlation between presence and reaction time hold when the interaction is manipulated to alter presence?

2 Background and Related Work

2.1 Measuring Presence in MR

Our definition of MR aligns with Augmented Reality (AR) on Milgram’s reality-virtuality spectrum [39], where virtual elements merge seamlessly with the physical world. The core of this experience is “presence,” the feeling of “being there,” which relies on interactions that reflect real-world behavior [50]. Accurately measuring *presence* in any VE (AR/VR/MR) presents unique challenges. Traditional post-experience questionnaires [71, 52, 51, 19,

59, 47, 49, 16], often used to measure presence, capture retrospective impressions of sensory fidelity and engagement but can lack real-time insights and may suffer from memory biases [53]. Originally designed for fully immersive VR [63], questionnaires have been adapted for AR/MR [64, 69] but still primarily assess peripheral factors rather than intrinsic presence [18, 44, 63, 48, 62]. Despite limitations, they are useful across the reality-virtuality spectrum [67].

Given the limitations of subjective measures, researchers have studied behavioral and physiological metrics like facial expressions [60], posture [17], heart rate [28], and skin conductance [37] as objective measures for presence. Still, these often yield inconsistent results [54, 53, 49]. Objective metrics, such as reaction time—the time taken to respond to a stimulus—have shown potential in measuring presence [33, 59, 4, 5, 58, 23, 6]. Recent research by Chandio et al. [12, 13] has demonstrated a correlation between presence and reaction time in MR environments: as presence decreases, reaction time increases. Particularly when influenced by technical elements such as place and plausibility illusions [71, 57] or by conditioning-related human factors, including cognitive perception and prior experiences [68, 29].

2.2 Presence and Interaction

Despite evidence suggesting that interaction design influences presence, research on the impact of interaction has yielded mixed results. While some studies found that high-fidelity interactions enhance immersion and enjoyment [65, 40, 22, 32, 35, 56], others reported similar outcomes between high- and low-fidelity interactions [47, 66]. Others have either focused on the sensor side of interaction [40] or explored interaction modalities [34], but none have directly addressed the impact of presence. Furthermore, many of these studies have only examined the interaction effect incidentally, treating it as a secondary research question rather than a central focus. These inconsistencies underscore the need for dedicated research to understand the relationship between interaction and presence comprehensively.

2.3 Interaction, Presence and Reaction time

While research has explored technical and human factors influencing presence and its relationship with reaction time, the direct link between presence, interaction, and reaction time requires more attention. Interaction plays a vital role in shaping how presence and reaction time relate. Interaction is crucial because it engages the user and directly triggers their reaction to cues. Users engage with VE through different interaction mechanisms like controllers, hand gestures, or voice commands, and these interactions significantly affect their sense of presence and responsiveness [42].

For instance, familiar tasks like manipulating and selecting objects foster a sense of control and agency, helping users engage with the virtual world naturally. Immediate and consistent feedback through visual, auditory, or haptic cues reinforces user actions and strengthens the presence [40]. However, the relationship between interaction, presence, and reaction time is not without its challenges. Overly complex or unfamiliar controls can slow reaction time and reduce the sense of presence [31]. Furthermore, task familiarity is crucial while designing the interaction [25, 10], as daily activities like moving and pointing need to be intuitive and consistent with users' mental models to enhance presence [41, 15] but may add latency to the reaction time.

3 Approach

3.1 Overview

The correlation between presence and reaction time in MR, mainly when influenced by different interactions, presents an exciting area of research. In Figure 2, we present a structured model that depicts how interaction types and tasks in MR affect presence and reaction

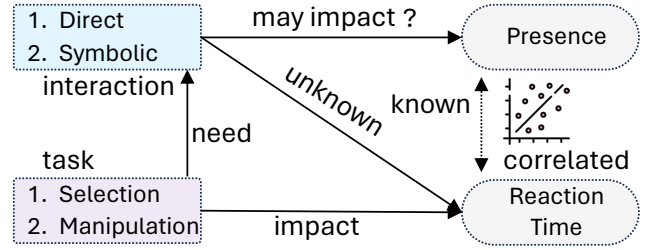


Figure 2: Model depicting the relationship between interaction types, tasks, presence, and reaction time.

time. Both of these are measurable quantities that may vary with different interaction-task combinations. The combination of these components forms the basis for the investigation of this paper. Now, we formally define the key elements of the model below:

Reaction Time. is the duration a user takes to respond to a cue.

Presence. is the psychological sensation of “being there” within the VE [67]. A sense of presence is not just about the high-fidelity graphics but about a well-designed VE that closely replicates real-world experiences.

Interactions. are the methods and mechanisms through which users can manipulate virtual objects or engage with the VE. This includes direct physical hand gestures, voice commands, controller inputs, and eye tracking, which facilitate a communication loop between the user and the VE. A key factor in designing VE is determining how interaction fidelity¹ supports high level goals of VE [8]. Interaction can be categorized in various ways [8, 24, 38, 56], such as single-handed versus two-handed or gesture-based versus controllers, near vs far interactions and so on. We focus on the high-level design of the interaction and classify that as follows:

Direct Interaction (D) is when the user engages with the VE in a manner that closely resembles real-world behavior. For instance, reaching out to grab a virtual object with a hand gesture, like grasping a physical object. These interactions are considered more natural and may create a heightened sense of presence because they align with the user’s natural, familiar movements and expectations.

Symbolic Interaction (S) refers to actions that do not directly replicate real-world action but instead rely on symbolic or abstract representations to achieve the desired result. For instance, using a button to select a virtual object instead of touching it. Although less intuitive, these interactions are still effective in enabling users to interact with the VE. Although D interactions are considered ideal due to their realism [7], they require more time and cost to develop. Symbolic interaction can sometimes be just as effective [43].

Tasks. represents a specific activity or goal users aim to accomplish in VE. It is an action-oriented step or process that the user undertakes while interacting with VE. Bowman et al. [11] categorized interaction techniques into selection, manipulation, travel, and system control tasks. [30] expanded on these by introducing canonical manipulation tasks: selection, positioning, rotation, and scaling. They highlighted the importance of using low-level tasks to evaluate interactions’ impact. The following tasks are chosen based on their prevalence and relevance in everyday virtual interactions [7, 27].

Selection task involves picking out or highlighting specific objects in VE. For instance, clicking on or tapping an object to mark it as selected or initiating an actuation. The simplicity of the task usually means it requires less effort and is less physically demanding than other tasks.

Manipulation is a more intricate task that requires reaching out and manipulating virtual objects, such as grabbing, moving, or rotating items in VE. This type of task demands higher interaction fidelity

¹Interaction fidelity measures how accurately real-world interactions are replicated [35].

than selection tasks due to the complexity of the movements involved, which makes it important for our evaluation.

The relationship between presence and reaction time in MR may be influenced by interactions and tasks in ways that are unexplored. So, what is known and unknown in this construct motivates further investigation into this relationship.

Known. Some form of user interaction is required to complete any task in the VE. Each task has a measurable reaction time that provides insights into user performance. Prior research has established a proven correlation between presence and reaction time. Literature also supports the idea that interactions may affect presence. For instance, direct interactions that mimic real-world behavior can enhance presence, while symbolic or abstract interactions may be less effective.

Unknown. It remains unclear precisely how different types of interactions affect reaction times. The context of how interaction influences user performance needs further exploration. The connection between specific combinations of interactions and tasks and how they impact the correlation between presence and reaction time is yet to be fully understood. This gap includes questions like: How do different tasks modify the influence of various interaction types on presence and reaction time? What patterns emerge from specific combinations of tasks and interactions that influence reaction time?

Factors affecting Reaction Time. Figure 3 aims to break down the factors influencing reaction time (RT) in VE to design the study to learn the sensitivity of the relationship between presence and RT to interaction. The main focus of Figure 3 is explaining the different RT contributors. Technical factors represent the quality and realism of VE and impact how quickly users react, as more realistic environments can reduce the mental load on users [70]. Human Factors are attributes that relate directly to the user and their physiological and psychological state. For example, the user's familiarity and prior experience with the VE affect their ability to respond quickly and effectively to stimuli. Combined factors consider the interactions between the human and technical aspects, such as interaction mechanisms. This affects how naturally a user can engage with VE, some mechanisms might require complex gestures or symbolic inputs that could slow down reaction times. The effectiveness of the interaction mechanism is crucial, as it directly contributes to how the VE appears and behaves and how the user will respond. Users build a mental model of interaction patterns based on expectations and past experiences [41]. Any deviation from expected behavior can affect presence; for instance, natural interactions, such as grabbing objects with realistic hand gestures, may lead to greater presence. In contrast, awkward or inconsistent interactions may disrupt presence, making users more aware of the virtual world's artificial nature. Expectations around the realism and consistency of interactions are crucial for user presence in VE. When interaction aligns closely with expectations (D), it fosters a sense of flow that allows users to interact intuitively with the VE. Conversely, symbolic interactions create a mismatch that may reduce presence. Lastly, other factors may also influence RT, such as task complexity, cognitive load, and environmental distractions; exploration of these elements is out of the scope of this paper. Due to interaction's potential influence, investigating reaction time for a given task and interaction pair and its variation is essential to understanding the correlation between presence and reaction time.

3.2 Hypothesis

Given these observations, we hypothesize that the interaction mechanisms in MR can affect the presence and their reaction times to stimuli. Our hypotheses (**H**) concerning interactions are as follows:

- H1:** Manipulating interactions leads to change in presence.
- H2:** Participants will have a higher sense of presence when experiencing direct interactions than symbolic interactions.

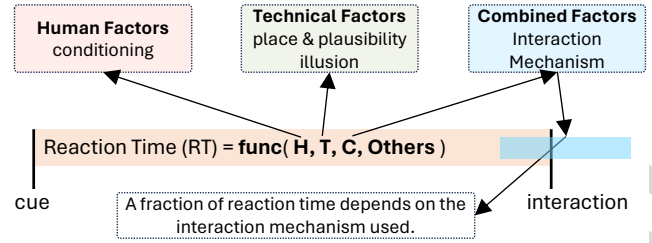


Figure 3: Reaction Time Decomposed.

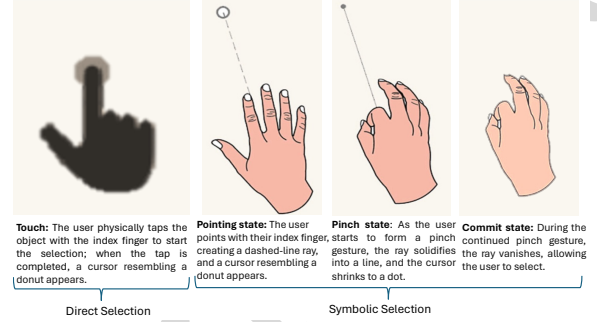


Figure 4: Hand gestures for selection task

- H3:** The effect of interaction type on presence is moderated by the type of task the user performs.
- H4:** The change in interaction type will lead to a change in presence, irrespective of the type of task.
- H5:** The change in interaction type will lead to a change in reaction time, irrespective of the type of task.
- H6:** The reaction time magnitude depends on the interaction type.
- H7:** Presence and reaction time are correlated.

3.3 Experimental Design

Selection Task (S) is designed as a color-matching bubble pop activity, where participants are required to match a bubble's color to a highlighted box among four options (red, green, blue, brown). We created a sequence of four color patterns, using a Latin square design to order the bubble colors repeatedly throughout the task. To determine which box to highlight, we employed the reverse diagonal values from our four color patterns matrix, ensuring that each color box was highlighted systematically and predictably across multiple trials. In the direct selection (DS) condition, participants are instructed to physically tap on the bubble directly with their index finger to pop it (make a selection as shown in Figure 4(left)) when its color-matched the highlighted box. Upon tapping, a donut-shaped cursor appears to confirm the selected object. This gesture replicates how users interact with bubbles in the real world.

In the symbolic selection (SS) condition, when the color of the bubble matches the highlighted box, the participants are asked to perform the airtap interaction [1] on the highlighted box to pop the bubble, as shown in Figure 4(right). The participants point toward the box using their index finger, which creates a ray depicted as a dashed line. A donut-shaped cursor appears on the box being pointed at, signifying where the participant is aiming. Then, the participants move their index finger toward their thumb, forming a pinch gesture. As the participants transition to a pinch gesture, the dashed ray changes to a solid line, and the donut cursor shrinks to a dot, indicating that they are preparing to make a selection. Once the pinch gesture is sustained, the solid ray disappears entirely, and the participants can select the box. The selection is finalized when the pinch gesture is complete. The placement of the bubble and

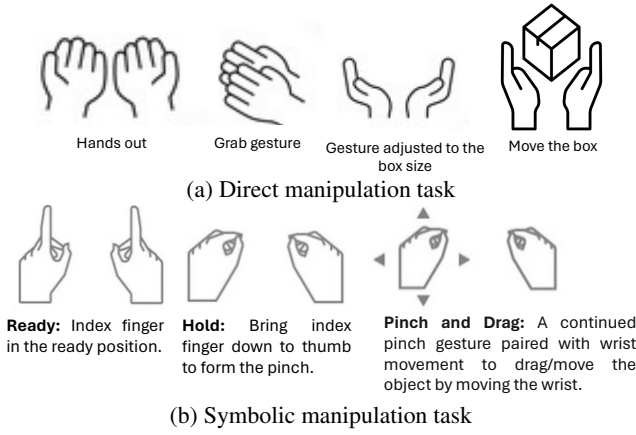


Figure 5: Hand gestures for the manipulation task.

boxes and interaction in SS is carefully designed to ensure that both the tasks in S have the same physical workload and arm movement. Although the box is farther than the bubble, the intentional ray formation of the line of sight with the box compensates for the delta in the distance. We use Fitts's law [20] to calculate the expected time of motor movement for both positions and corroborated that empirically during our pilot study (§4.5).

Manipulation Task (M) For the manipulation task, participants are presented with a virtual box positioned in front of them, which they must move back and forth each time the box turns green. In the direct manipulation (DM) condition, participants performed a natural two-hand grab gesture to interact with the box, simulating real-life object movement, as shown in Figure 5(a). Participants start by positioning their hands in front of them, palms facing up, and extending their fingers outward to prepare for the gesture. Participants are then asked to flex their fingers inward, imitating a grab gesture. They adjust this gesture to match the box size, creating a realistic grip that ensures accurate interaction. Once a secure virtual grip was established, users could reposition the box by moving their wrists and arms. The box followed the hand movement, allowing users to relocate it as needed.

For the symbolic manipulation (SM) condition, we implemented a bounding box around the virtual object. We highlighted a specific area (designed as a 'hook'), avoiding needing a full two-hand grab gesture. As illustrated in Figure 5(b), the participants hold their index finger in a pointing position to the hooks on each side of the box, slightly elevated and ready for action. This starting posture ensures accurate hand tracking by the system. The participants move their index fingers toward their thumb, forming a pinch gesture. The thumb and index finger approach each other but do not necessarily make contact to indicate the intent to interact with a box—the hold phase transitions from mere pointing to active interaction. Once the pinch gesture is initiated, the participants maintain it while moving their wrists, effectively dragging/moving the box in the desired direction (back and forth).

3.4 Interaction Implementation

Interactions in this study are managed through the HoloLens' built-in hand tracking (accuracy of HoloLens2 hand tracking is empirically tested by [26, 61]), allowing participants to use their hand's gestures to move the box (M) and pop bubble (S), as they would in the physical world. Pointing and clicking gestures are excluded from the D conditions to maintain realism. Each gesture step flows into the next with tactile feedback (cursor or ray or color change) to the participants and to ensure that gestures are interpreted correctly by the system. Following HoloLens 2's near-interaction guidelines, primary objects (boxes and bubbles) are placed 35-50 cm from the participant's abdomen for ease of access and are ergonomically positioned to prevent muscle fatigue [21]. To reduce accidental selec-

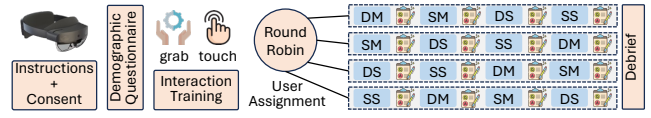


Figure 6: User study timeline consisting of pre-experiment procedures and users performing all four experimental conditions in one of the four orders devised using Latin square order. The four experimental blocks include DM, SM, DS, SS.

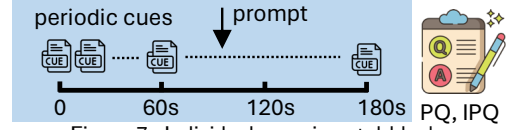


Figure 7: Individual experimental block.

tions, participants are taught specific hand gestures consistent with the standard HoloLens 2 hand gestures interaction paradigm [3]. Lastly, the box is placed on the table surface in M conditions to compensate for the haptic feedback.

4 User Study

This section presents the details of our user study, experimental conditions, measures, and procedures.

4.1 Participants

We conducted an a priori power analysis to determine the appropriate sample size for our within-subject four conditions (DM, SM, DS, SS) study for medium size effect [14] with four repeated measures and nonsphericity correction, the analysis recommended a minimum of 46 participants. We recruited 50 participants to account for attrition (33 male-identifying, 17 female-identifying) with a mean age of 21.86 years (ST = 3.39). All participants volunteered and provided written informed consent. They received \$15 for their participation. All the participants had normal or corrected normal vision with contact lenses or glasses. The demographic distribution and media usage of participants can be seen in Table 1. The institution's ethics committee approved the study.

4.2 Material

We selected HoloLens 2 [2] due to its direct view of the real world, a self-contained device with a holographic processing unit with a Qualcomm Snapdragon 850 CPU, featuring eye, spatial, and hand-tracking capabilities and dual displays with a 1440 × 936 pixels resolution with 110°FoV. The VEs were developed using the Unity game engine and C# programming language with the Unity XR Plug-in Management package, optimized for the Universal Windows Platform. Custom scripts were added to facilitate hand tracking and reaction time recording.

4.3 Experimental Task

In Section §3.3, we explain the experimental conditions and designs. Here, we will summarize the physical tasks (move, touch, grab, point), virtual object placements, and how they are ordered in the study session. The study features four conditions to evaluate two interactions across two tasks (as shown in Figure 1). For two S task conditions, we designed a color-matching activity involving a bubble and four boxes. Participants stand as the scene is rendered before them, with bubbles positioned at their eye level and adjusted for height. Every 3s (seconds), the bubble changes color, and every 9s, the box color changes. Participants must pop the bubble when its color matches the box's, completing 20 bubble-popping trials in 3 minutes. The delay and duration were empirically adjusted during the pilot study (§4.5). In the D interaction, participants pop bubbles by directly touching them, while in the S interaction, they use an air-tap gesture to interact with the boxes. See §3.3 for details.

Table 1: Demographic data and media usage (past 5 years) across all conditions and participants. The key for frequency: never/almost never; rarely (< 2times); occasionally (a few times); frequently in the past; frequently (> 2times/month).

demographics	# participants
gender	17 female; 33 male
age	mean = 22.7 years (STD = 4.43)
frequency of VR experience	10 never used; 10 rarely; 20 occasionally; 5 frequently; 5 frequently in the past
frequency of AR/MR experience	17 never used; 17 rarely; 12 occasionally; 4 frequently; 0 frequently in the past
frequency of Gaming	5 never used; 3 rarely; 12 occasionally; 25 frequently; 5 frequently in the past

For two M task conditions, we designed a box moving activity using a single virtual box. Participants are asked to sit in a chair positioned in front of a table and adjust the chair to ensure they can comfortably reach the box. Every 5 seconds, the box turns green, signaling the participant to move the box to the side. Starting from a neutral position, they move the box to the right in one trial, then to the left in the following trial, continuing this alternating pattern for 3 minutes across 35 trials. To prevent participants from moving the box too far, white tape marks were pasted on the table 8 inches apart to indicate the target areas. The distance of movement was empirically determined during the pilot study. In the D interaction, participants use both hands to move the box with a real-world mimicking grab gesture directly. In contrast, in the S interaction, they use a pinch gesture to manipulate hooks on the box (§3.3).

In all conditions, we measure the *reaction time* as an interval between the time the color change occurred (cue) and the actual action taken by the participant (interaction). The feedback is explicit in all conditions: successful actions are marked by either the bubble popping or the box changing position, depending on the specific condition. To build a comprehensive response time profile, we concurrently recorded timestamps of all events, including color changes, touch, movement, and transitions, in a separate thread. Additionally, we logged the distance each participant moved the box, identified which box was moved, and verified if it matched the highlighted box. This comparison determined the accuracy: a perfect match indicated the correct color pairing. While memory retention is not a focus of this study, we log this data to exclude response times for incorrect color matches, ensuring comparability across all participants and conditions.

4.4 Measurements

We measured presence using two post-experience questionnaires: the Witmer and Singer Presence Questionnaire (PQ)[71] and the Igroup Presence Questionnaire (IPQ)[52, 51]. The PQ evaluates factors such as the possibility to act and examine, realism, self-evaluation, and interface quality. The IPQ measures three factors: the sense of physically being in the VE, realism, and involvement (crucial for evaluating interactions). The presence scores were derived from 33 items (14 IPQ and 19 PQ) on a 7-point Likert scale, with no modifications to the original questions. Additionally, we included a prompt question, "How connected do you feel to this virtual world?" on a 7-point scale, created by [13], to measure presence within the scene between trials in all four conditions and to align with reaction time. Our software recorded reaction time on the HoloLens 2 in milliseconds (ms). We collected 20 reaction time measurements for each participant for the S conditions and 35 for the M conditions.

4.5 Pilot Study

Before the main study, we ran a qualitative pilot with five participants to fine-tune the task parameters to minimize biases and enhance study reliability. By employing a think-aloud protocol, we collected insights on participant comfort and observations without

steering preferences toward specific parameters. This included testing different intervals for color changes (2, 3, 5, 10, and 15 seconds) and session duration (2 to 10 minutes). We also explored varying distances for box placement (5 to 20 inches) and assessed participant feedback on physical discomfort, color visibility, and object transparency. Feedback indicated a preference for a consistent size scale of boxes and bubbles; transparent bubbles were generally less favored. The pilot study's data helped refine the main study's design, ensuring tasks were ergonomically viable and technologically functional. Using the desktop application to establish consistent baseline times across all conditions, we assessed reaction times and accuracy (color matching) in S conditions.

4.6 Procedure

The study procedure is shown in Figure 6. Participants began by reviewing and signing the consent form, then providing demographic information like gender, age, and familiarity with AR/VR/gaming. They were briefed on the study conditions (experiments and questionnaires), scene applications, headset features, interactions to be performed (instruction on how to perform), appropriate hand poses, prompt questions, visual stimuli, actions they need to perform, and feedback they will receive as a result of successful interaction. In the briefing, participants are primed to answer correctly and not hastily by iterating that they need to focus on the correctness of the task in each condition.

In within-subjects designs, managing order effects is essential to measure the experimental conditions' impact accurately. To handle this, we used a balanced Latin square design [46] to randomize the order of the four conditions across the participants. Each condition was placed in each position an equal number of times and directly followed others equally. The 4x4 Latin square pattern, repeatedly multiple times in a round-robin manner, ensured proper distribution: {DM, SM, DS, SS}, {SM, DS, SS, DM}, {DS, SS, DM, SM}, {SS, DM, SM, DS} This approach ensured each condition occupied every position once per set of four participants.

In all conditions, participants wore the headset and performed two successful interactions (bubble popping and box moving) to familiarize themselves before each condition. These practice interactions were recorded but excluded from the analysis. As shown in Figure 7, periodic cues are generated (3s in S and 5s in M), and participants answered a prompt question halfway through each condition (1.5 minutes in). Participants removed the headset after three minutes, completing all trials (35 in M conditions and 20 in S conditions) and answering the prompt question. After each condition, they filled out presence questionnaires and responded to an open-ended question. After completing all four conditions, we conducted a small debriefing session where we answered their questions. The study session, including the initial briefing, four conditions, questionnaires, and debrief, took about 45 minutes.

5 Results

This section presents the results from our experiments using a within-subject study design when measuring presence using presence questionnaires (subjective) and reaction time (systematic). We conducted our analysis using the *scipy* library in Python.

5.1 Statistical Analysis

Table 2 summarizes the descriptive results from our statistical analysis, where we aggregated presence scores from questionnaires, prompt scores, and reaction time values. For the PQ and IPQ presence questionnaires, we reported values for all of the subscales. We report results for all experimental scenarios separately.

We used Levene's homogeneity test, which held true for all dependent variables. The Shapiro-Wilk test found that the aggregate questionnaire, individual questionnaires, prompt score, and reaction time were normally distributed. Among the different subscales, self-evaluation of performance (EVAL), possibility to exam-

Table 2: Summary of descriptive results for presence score using PQ and IPQ questionnaires, prompt score, and reaction time across experiments: direct selection (DS), symbolic selection (SS), direct manipulation (DM), and symbolic manipulation (SM). **All entries are given as mean value (standard deviation).** The subscales for PQ and IPQ questionnaires are Possibility to act (ACT), interface quality (IFQUAL), realism (REAL), possibility to examine (EXAM), self-evaluation of performance (EVAL), general presence (GP), spatial presence (SP), and involvement (INV).

Exp.	PQ (1-7)					IPQ (1-7)				Prompt	Reaction Time
	ACT	IFQUAL	REAL	EXAM	EVAL	GP	SP	INV	REAL	(1-7)	(seconds)
DS	5.35 (1.07)	5.05 (1.19)	4.91 (1.34)	5.18 (1.34)	5.50 (1.42)	5.00 (1.30)	4.46 (0.79)	4.22 (0.96)	4.22 (0.97)	5.05 (1.05)	1.065 (0.43)
SS	4.27 (1.10)	4.15 (1.22)	3.85 (1.01)	4.89 (1.24)	5.24 (0.96)	4.92 (0.91)	4.19 (0.62)	3.70 (0.69)	3.50 (1.19)	4.55 (1.30)	1.824 (0.21)
DM	5.91 (0.74)	5.55 (0.99)	5.50 (0.83)	5.75 (0.94)	6.25 (0.94)	5.40 (1.36)	4.18 (0.45)	4.80 (0.68)	4.62 (0.84)	5.23 (1.24)	1.915 (0.54)
SM	4.55 (0.96)	4.57 (0.91)	4.43 (0.80)	5.59 (1.04)	6.30 (0.75)	5.38 (1.39)	4.08 (0.52)	4.11 (0.62)	3.92 (1.18)	5.08 (1.10)	2.215 (0.71)

ine (EXAM), and general presence (GP) failed the normality test, while other subscales were found to be normally distributed. We do not evaluate the impact of our dependent variables on the three aforementioned subscales, in addition to spatial presence (SP), as summary statistics show that interaction type and task type did not significantly impact these subscales.

Table 3 shows the results for two-way repeated measures analysis of variance (ANOVA) with combined questionnaire scores, individual subscores, prompt score, and reaction time as the dependent variables. We used standard ANOVA as the possibility to act (ACT), interface quality (IFQUAL), realism (REAL) for both PQ and IPQ, and involvement (INV) subscales were found to be normally distributed. The interaction type and task type had a statistically significant impact on the questionnaire-based presence scores and reaction time. The effect of interaction type had a higher impact than interaction type across all presence scores, but the task type had a higher impact on reaction time. The effect of interaction and task types on prompt scores was barely statistically significant.

The cross effect was not observed for any dependent variable except for the reaction time. However, the effect was very small and could explain a very small variance in reaction time. Finally, in Figure 8, we visualized the relationship between presence scores and reaction times and quantified the correlation between the two measures, which was found to be modest (-0.42 to -0.63). The figure at the top also includes the linear regression lines expressing the relationship between the two measures.

5.2 Control Measures

The participants' familiarity with AR/MR, VR, and video games was not evenly distributed, as shown in Table 1. Due to this uneven distribution, an extensive statistical analysis was not possible, and these control measures were not considered further.

As discussed in the study procedure (Section 4.6), the participants were assigned one of the four experiment orders based on the Latin square pattern to manage the order effect. To confirm that the experiment order did not impact our outcomes, we conducted ANOVA of presence score, prompt score, and reaction time against the experiment order. To do that, an aggregate value using the sum across all experiments was computed for each user. The four groups of users were compared using experiment order as the independent variable. Our results suggested that experiment order did not significantly impact the presence scores ($F=0.79$, $p=0.54$), prompt score ($F=1.34$, $p=0.13$), and reaction time ($F=2.513$, $p=0.09$).

5.3 Interpretation of Results

In this section, we interpret the effects identified in our statistical analysis and explore the reasons for their manifestation. Our study used the overall change in presence as the interaction mechanism changed as the first check; the relationship between presence and reaction time would not be observed if the presence did not change. We observed a significant difference in overall presence, which was found to be statistically significant (Table 3). Therefore, we accept **H1**. We also found that the change in the interaction mechanism changed the overall presence, and the individual subscales showed

Table 3: Summary of the ANOVAs' results using questionnaires and reaction time as a presence measure.

Presence Measure	Factor	F-stat	p-value	η^2
Questionnaire				
Main effects	interaction	57.38	<0.05	0.207
	task	22.57	<0.05	0.082
Cross effect	interaction:task	0.002	0.967	0.002
ACT				
Main effects	interaction	78.79	<0.05	0.276
	task	9.927	<0.05	0.035
IFQUAL				
Main effects	interaction	36.72	<0.05	0.152
	task	8.794	<0.05	0.036
PQ-REAL				
Main effects	interaction	53.85	<0.05	0.203
	task	15.98	<0.05	0.060
INV				
Main effects	interaction	31.93	<0.05	0.128
	task	21.37	<0.05	0.085
IPQ-REAL				
Main effects	interaction	22.00	<0.05	0.098
	task	7.307	<0.05	0.032
Prompt Score				
Main effects	interaction	3.899	0.0497	0.019
	task	4.591	0.0333	0.022
Cross effect	interaction:task	1.103	0.2948	0.005
Reaction Time				
Main effects	interaction	54.43	<0.05	0.163
	task	74.62	<0.05	0.223
Cross effect	interaction:task	1.028	0.016	0.003

consistent statistical significance. The interaction mechanisms differed in enabling users to act, the quality of the interface, how realistic they felt, and the level of involvement they needed to work. As a result, the interaction type was the dominant independent factor that significantly impacted the presence score across subscales. Furthermore, as the direct interactions were more involved than the symbolic ones, the presence score for DS and DM was higher (Table 2) and statistically significant, and **H2** can be accepted.

The different tasks differed in the level of involvement they expected from the participant; it was expected that changing a task would impact presence. However, we also expected that the nature of the task might impact the effect of interaction type on presence; for example, the participant may find a given interaction mechanism better suited to one task than the other. However, our results suggested that while task impacts presence, the effect of the two interaction types on presence does not change. We did not observe an interaction effect at the overall questionnaire and subscales levels (omitted for brevity). Therefore, **H3** is rejected. Our next hypothesis took a more specific approach to the cross-over between task and interaction, stating that if the interaction changes, there will be

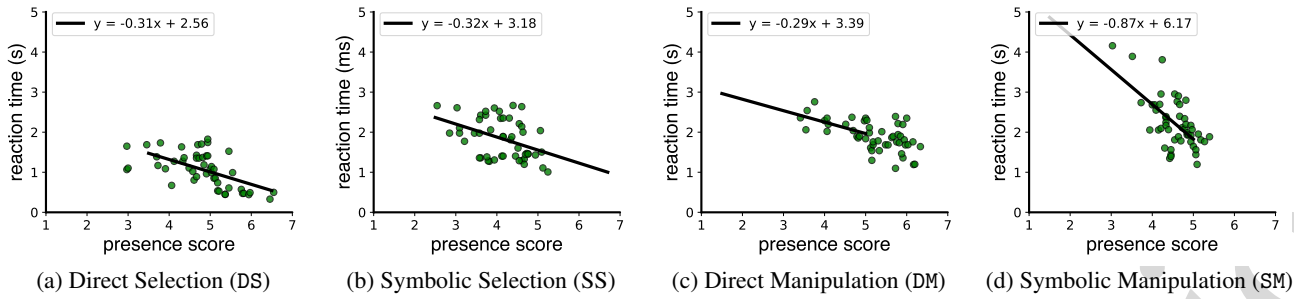


Figure 8: *Presence vs. Reaction Time: Presence decreases as reaction time increases. Reaction time and presence also show a modest correlation: overall (-0.54), DS (-0.59), SS(-0.42), DM (-0.59), and SM (-0.63). Each green circle represents a study participant. The black line is the linear regression fit for the data.*

a change in presence. This was observed, and **H4** can be accepted.

The first four hypotheses focused on the relationship between presence score, task type, and interaction type. We next looked at the interplay between these factors and reaction time. Understandably, reaction time is naturally impacted by the type of interaction. For example, a simple touch interaction would take a lot less time than a manipulation interaction that requires a more careful analysis of the virtual and physical worlds. However, the reaction time is not just the function of the interaction type, but also the task type. Therefore, it was important to analyze that the effect of interaction type is consistent irrespective of the task type. As shown in Table 3, the impact of interaction type on the reaction time was found to be statistically significant and **H5** is accepted.

Additionally, similar to its effect on presence, the task type impacted reaction time as well. However, interestingly, the reaction time was the only measure and the dependent variable where we observed the cross effect between interaction type and task type. In presence questionnaires, the scale is the same (0–7), and the effect of interaction explains most of the variance in the observed score. However, in reaction time, both tasks take different amounts of time, and therefore, a significant variance is explained by the task type in addition to the interaction type. In our statistical analysis, the effect of task type also shows that these systemic measures, such as reaction time, depend on the task at hand in addition to the user's level of presence. As a result, they demonstrate a cross-effect in addition to the main effects. Therefore, **H6** is accepted.

Finally, as a causal relationship between presence and reaction time cannot be established, we look at the correlation between the two presence measures across different experiment types in our study. As shown in Figure 8, a moderate to high correlation between the presence score and the reaction time is observed for the overall scores as well as across all the individual experiments. Therefore, **H7** stating a negative correlation between the level of presence and reaction time is accepted.

5.4 Thematic Analysis

While our quantitative results demonstrate that different conditions affected presence scores and reaction times, we wanted to investigate the factors contributing to these changes. To do this, we disaggregated the answers to the open-ended question and subscales: We do not present any results for general and spatial presence-related comments as they are highly correlated with overall scores.

Involvement: As both quantitative and qualitative results suggested, transitioning between different experimental conditions often caused participants to feel a heightened sense of involvement. Comments such as "Being able to control the VE directly was engrossing" and "I felt really part of the virtual world when the interactions were responsive" were common in DM and a few in DS.

Realism: Similarly, the shift to conditions DM and DS was noted by the participants who commented, "The virtual objects looked almost tangible," and "The realistic response of the environment to

my actions enhanced the fun". This suggests that visual and interaction realism enhancements contribute to the sense of presence.

Interface Quality: Feedback on interface quality often highlighted the limitations of the field of view and control responsiveness. For example, one participant noted, "The small field of view limited my interaction," while another said, "Sometimes the gestures did not register, which pulled me out of the experience" (reported in SS). These comments align with observations where interface limitations negatively impacted user experience.

Comfort: The physical comfort of using the headset also played a crucial role in the presence experience. Participants frequently mentioned comfort issues, stating, "The headset felt heavy after a while," or "I experienced some eye strain during the last few trials."

The thematic analysis revealed consistent themes across conditions, especially regarding the impact of technical features on the presence. Enhancements in realism and interaction quality were positively received, while limitations in interactions and physical discomfort were areas of concern. These insights from thematic analysis are supported by the quantitative results from the subscales, providing a comprehensive understanding of the factors influencing presence in VE.

6 Discussion

Our findings revealed significant effects of interaction and task types on presence scores, as indicated by our two-way repeated measures ANOVA. The interaction type showed a higher influence on the presence scores across all subscales, while task type had a more pronounced effect on reaction time. Notably, the interaction between task and interaction type had a negligible impact, suggesting that the influence of interaction type on presence and reaction time is robust across different tasks.

Correlation between Presence and Reaction Time. The correlation between presence and reaction time was modestly negative, ranging from -0.42 to -0.63 across different scenarios. This correlation is visualized in Figure 8, with linear regression lines illustrating the relationship between these measures. The trend indicates that as presence increases, reaction time decreases, suggesting that higher levels of presence in an MR can lead to faster responses.

Interpretation of Hypotheses. H1 and H2 were supported, indicating that direct interactions, which are more immersive, consistently enhance the sense of presence compared to symbolic interactions. This is reflective of the intuitive nature of direct interactions that closely mimic real-world actions, facilitating a stronger engagement with the VE. H3 was not supported, which highlights that the superior efficacy of direct interaction in enhancing presence is not contingent on the task type. This suggests a general preference or suitability of direct interactions over symbolic ones in MR, regardless of the specific task being performed. H4 was confirmed, underscoring that changes in interaction type are sufficient to alter the sense of presence significantly. This finding is crucial for designers of MR interfaces, as it emphasizes the importance of choosing appropriate interactions to enhance the overall

experience. H5, H6, and H7 highlight the relationships between interaction types, reaction times, and presence. These results confirm that interaction types not only affect presence but also significantly impact task performance efficiency, as reflected in reaction times.

Theoretical and Practical Implications. The implications of these results are multifaceted. Theoretically, they contribute to the understanding of how different interactions influence user experience in MR, offering insights into the psychological and physiological aspects of presence. Practically, these findings can guide the development of more effective mixed reality applications, where the choice of interaction type can be tailored to enhance both the sense of presence and the efficiency of performance (reaction time).

7 Limitations and Future Work

In this section, we briefly discuss the limitations of using reaction time as a presence measure as well as our study. We also outline the potential future directions for this work.

Absolute vs. Relative Presence Measure. For the questionnaire score, a statement can be made about the level of presence a user is experiencing. However, since reaction time also depends on the task at hand, just looking at it does not provide enough information about the level of presence. At present, reaction time is best suited for measuring the change in presence rather than the absolute level of presence. Further studies should explore whether the effect of task type on reaction time can be modeled, as this may enable the use of reaction time as a standalone absolute measure of presence.

Future Directions. Further research could explore the long-term effects of these interactions on user adaptation and learning. Analyzing prolonged exposure to varied interactions will reveal how users adapt over time, potentially developing new strategies or behaviors. This research could uncover patterns in the correlation between sustained exposure and presence scores, and how extended exposure influences reaction times. Exploring multisensory feedback integration can significantly enhance immersion and presence. By incorporating auditory, haptic, and olfactory cues, researchers could examine how these additional sensory channels influence presence and reaction time relationship. VE that allow multi-user interactions provide a unique opportunity to study social presence. Researching how users perceive others in shared virtual spaces and how collaboration impacts overall presence and reaction time could be an interesting avenue.

8 Conclusion

In conclusion, this study provides a comprehensive analysis of how interaction and task types influence presence and reaction time, offering valuable insights for enhancing user experience in MR environments. The confirmation of our hypotheses strengthens the argument that direct interactions, which are inherently more immersive, enhance presence compared to symbolic interactions, reflecting the intuitive nature of direct manipulation in VE. The study confirms that changes in interaction type alone can meaningfully alter presence and reaction time. This finding has practical implications for MR developers, highlighting the importance of careful interaction designs to enhance both the sense of presence and performance efficiency. In summary, these findings expand our understanding of MR interactions and point to reaction time as a promising metric for assessing presence in future MR applications, warranting further exploration.

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